Slide 1 – Title

Good afternoon everyone.
We are **Md. Abdullah Al Muntasir, Md. Minhaj Uddin Raiyan, and Imran Yakub**.

Today, I will be presenting our project titled **"A Lightweight ANPR System."**

Slide 2 – Outline

This presentation will follow this outline.
First, I will briefly introduce ANPR and the motivation behind our work.
Then I will explain the objectives, methodology, results and analysis,
and finally conclude with key findings and future improvements.

Slide 3 – Introduction

ANPR, or **Automatic Number Plate Recognition**, is a computer-vision system that automatically detects and reads vehicle license plates from images.

It is widely used in **smart traffic systems, parking management, toll collection, and security applications**.

However, identifying number plates manually is inefficient for modern systems.
Traditional rule-based methods often fail when lighting conditions change, images are blurred, or plates are captured from different angles.

Therefore, an **automated recognition system** provides faster, more consistent, and more accurate results.

Slide 4 – Objectives

The objectives of this project are:

First, **to understand the fundamental image preprocessing steps** used in computer vision.

Second, **to detect the number plate region** from a vehicle image using contour-based localization.

Third, **to extract and crop the detected plate area** for further processing.

Fourth, **to perform optical character recognition**, or OCR, to read the alphanumeric text on the plate.

And finally, **to verify the system's accuracy** by testing on multiple images and analyzing the results.

Slide 5 & 6 – Methodology

Now I will explain the methodology of our system.

First, in **image acquisition**, the user uploads a vehicle image using the file upload function.

Next comes **preprocessing**.
The image is converted from BGR to grayscale to reduce complexity.
Then, a **bilateral filter** is applied to reduce noise while preserving edges.
After that, **Canny edge detection** is used to highlight strong edges such as number plate boundaries.

In **contour detection**, all contours are extracted from the edge-detected image.
The largest contours are analyzed, and polygon approximation is applied.
A four-sided contour is selected as the probable number plate.

Then comes **masking and cropping**.
A binary mask is created to isolate the plate region, and the plate is cropped tightly.

After that, **EasyOCR** is used to read characters from the cropped plate region.
The detected characters are combined into a single text string.

If contour detection fails, a **fallback strategy** is applied.
In that case, OCR is directly applied to the full image to avoid complete system failure.

Finally, the recognized plate number is displayed and highlighted on the original image.

Slide 7 – Result and Analysis

This slide shows the step-by-step output of our system.

Starting from the original image, it is converted to grayscale, noise is reduced, edges are detected, contours are identified, the plate is cropped, and finally the detected text is displayed.

We can clearly see that the system successfully detects the number plate and reads the characters accurately.

Slide 8 & 9 – Results, Observations, and Limitations

From our experiments, we observed that:

The system successfully detects the plate region in clear images.
EasyOCR recognizes alphanumeric characters with high accuracy when the plate is distinct.
The combination of bilateral filtering and Canny edge detection produces clean contours.

However, there are some limitations.

Accuracy decreases in **low-light conditions**.
Blurred or low-quality images can cause OCR errors.
The system does not correct tilted or rotated plates.
Complex backgrounds and non-standard fonts also reduce performance.

Slide 10 – Conclusion

To conclude, we successfully implemented a **lightweight ANPR system** using classical image processing and EasyOCR.

The system efficiently detects number plates and extracts characters without requiring deep learning models.

Compared to deep-learning-based ANPR systems, our approach is simpler, faster, and suitable for small-scale applications.

In the future, performance can be improved by integrating **YOLO-based plate detection**, adding **rotation correction**, and supporting **Bangla license plates**.

Slide 11 – Thank You

That concludes our presentation.
Thank you very much for your attention.
I will be happy to answer any questions.